

# 1 First Fundamental Form

• Let  $S : \mathbf{r} = \mathbf{r}(u, v)$  be a regular parametric surface, then for any point  $p = \mathbf{r}(u_0, v_0) \in S$ , we know that the tangent space  $T_p S$  at  $p$  is given by  $T_p S = \text{Span}_{\mathbb{R}}\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\}$ , where  $\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\}$  is a basis for the 2-dimensional vector space  $T_p S$  (we may also denote  $\partial_u \mathbf{r}(u_0, v_0) := \partial_u \mathbf{r}|_{(u_0, v_0)}$  and so on). The inner product of any two tangent vectors to the surface  $S$  at any point is defined to be their inner product as vectors in  $\mathbb{R}^3$ .

• For any point  $p = \mathbf{r}(u, v) \in S$ , we have a tangent vector

$$d\mathbf{r}(u, v) = \partial_u \mathbf{r}(u, v)du + \partial_v \mathbf{r}(u, v)dv \in T_p S,$$

where  $(du, dv)$  is the components of  $d\mathbf{r}(u, v)$  under the basis  $\{\partial_u \mathbf{r}(u, v), \partial_v \mathbf{r}(u, v)\}$ .

• In general,  $\{\partial_u \mathbf{r}(u, v), \partial_v \mathbf{r}(u, v)\}$  is not an orthonormal basis, so we wish to find the metric coefficients of the basis  $\{\partial_u \mathbf{r}(u, v), \partial_v \mathbf{r}(u, v)\}$ , so we denote

$$E(u, v) := \partial_u \mathbf{r}(u, v) \cdot \partial_u \mathbf{r}(u, v),$$

$$F(u, v) := \partial_u \mathbf{r}(u, v) \cdot \partial_v \mathbf{r}(u, v),$$

$$G(u, v) := \partial_v \mathbf{r}(u, v) \cdot \partial_v \mathbf{r}(u, v),$$

which yields a symmetric matrix

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} = \begin{pmatrix} \partial_u \mathbf{r} \\ \partial_v \mathbf{r} \end{pmatrix} \begin{pmatrix} \partial_u \mathbf{r} & \partial_v \mathbf{r} \end{pmatrix},$$

note that  $\det E > 0$  and

$$\begin{aligned} \det \begin{pmatrix} E(u, v) & F(u, v) \\ F(u, v) & G(u, v) \end{pmatrix} &= E(u, v)G(u, v) - F(u, v)^2 \\ &= |\partial_u \mathbf{r}|^2 |\partial_v \mathbf{r}|^2 - (|\partial_u \mathbf{r}| |\partial_v \mathbf{r}| \cos \langle \partial_u \mathbf{r}, \partial_v \mathbf{r} \rangle)^2 \\ &= |\partial_u \mathbf{r}|^2 |\partial_v \mathbf{r}|^2 (1 - \cos^2 \langle \partial_u \mathbf{r}, \partial_v \mathbf{r} \rangle) \\ &> 0. \quad (\text{since } \cos \langle \partial_u \mathbf{r}, \partial_v \mathbf{r} \rangle \neq 1) \end{aligned}$$

This means that the metric coefficient matrix  $\begin{pmatrix} E & F \\ F & G \end{pmatrix}$  is positive definite.

## Definition 1.(First Fundamental Form)

For a regular parametric surface  $S : \mathbf{r} = \mathbf{r}(u, v)$ , define

$$\begin{aligned} I &:= d\mathbf{r}(u, v) \cdot d\mathbf{r}(u, v) \\ &= (\partial_u \mathbf{r}(u, v)du + \partial_v \mathbf{r}(u, v)dv) \cdot (\partial_u \mathbf{r}(u, v)du + \partial_v \mathbf{r}(u, v)dv) \\ &= E(u, v)(du)^2 + 2F(u, v)dudv + G(u, v)(dv)^2 \\ &= \begin{pmatrix} du & dv \end{pmatrix} \begin{pmatrix} E(u, v) & F(u, v) \\ F(u, v) & G(u, v) \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}, \end{aligned}$$

which is called the **first fundamental form** of  $S$ .

**Remark:** From the equation

$$d\mathbf{r}(u, v) = \partial_u \mathbf{r}(u, v) du + \partial_v \mathbf{r}(u, v) dv \in T_p S,$$

$(du, dv)$  should be viewed as the components of  $d\mathbf{r}(u, v)$  under the basis  $\{\partial_u \mathbf{r}, \partial_v \mathbf{r}\}$ , so  $du, dv \in (T_p S)^*$ , where we use the notation  $V^* := \text{Hom}(V; \mathbb{R})$  to denote the space of linear functionals on  $V$ . So from this viewpoint, we see that the first fundamental form on  $S$  is actually the inner product on each tangent space  $T_p S$  inherited from  $\mathbb{R}^3$ , which is a quadratic form.

**Proposition 1.** Let  $S : \mathbf{r} = \mathbf{r}(u, v)$  be a regular parametric surface, for any admissible change of parameters

$$u = u(\tilde{u}, \tilde{v}), \quad v = v(\tilde{u}, \tilde{v}).$$

then

$$I = \tilde{I},$$

where  $I = E(du)^2 + 2F du dv + G(dv)^2$  and  $\tilde{I} = \tilde{E}(d\tilde{u})^2 + 2\tilde{F} d\tilde{u} d\tilde{v} + \tilde{G}(d\tilde{v})^2$ . i.e., the first fundamental form is invariant under any admissible change of parameters.

**Proof:** Since  $S : \mathbf{r} = \mathbf{r}(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v})) \triangleq \mathbf{r}(\tilde{u}, \tilde{v})$ , thus we get

$$\begin{aligned} d\mathbf{r} &= \partial_u \mathbf{r}(u, v) du + \partial_v \mathbf{r}(u, v) dv \\ &= \partial_u \mathbf{r}(u, v) \left( \frac{\partial u}{\partial \tilde{u}} d\tilde{u} + \frac{\partial u}{\partial \tilde{v}} d\tilde{v} \right) + \partial_v \mathbf{r}(u, v) \left( \frac{\partial v}{\partial \tilde{u}} d\tilde{u} + \frac{\partial v}{\partial \tilde{v}} d\tilde{v} \right) \\ &= \partial_{\tilde{u}} \mathbf{r}(\tilde{u}, \tilde{v}) d\tilde{u} + \partial_{\tilde{v}} \mathbf{r}(\tilde{u}, \tilde{v}) d\tilde{v}. \end{aligned}$$

Moreover, we obtain

$$\begin{cases} \partial_{\tilde{u}} \mathbf{r}(\tilde{u}, \tilde{v}) = \partial_u \mathbf{r}(u, v) \frac{\partial u}{\partial \tilde{u}} + \partial_v \mathbf{r}(u, v) \frac{\partial v}{\partial \tilde{u}}, \\ \partial_{\tilde{v}} \mathbf{r}(\tilde{u}, \tilde{v}) = \partial_u \mathbf{r}(u, v) \frac{\partial u}{\partial \tilde{v}} + \partial_v \mathbf{r}(u, v) \frac{\partial v}{\partial \tilde{v}}. \end{cases}$$

and

$$\begin{cases} du = \frac{\partial u}{\partial \tilde{u}} d\tilde{u} + \frac{\partial u}{\partial \tilde{v}} d\tilde{v}, \\ dv = \frac{\partial v}{\partial \tilde{u}} d\tilde{u} + \frac{\partial v}{\partial \tilde{v}} d\tilde{v}. \end{cases}$$

i.e., we get

$$\begin{pmatrix} \partial_{\tilde{u}} \mathbf{r} \\ \partial_{\tilde{v}} \mathbf{r} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial \tilde{u}} & \frac{\partial v}{\partial \tilde{u}} \\ \frac{\partial u}{\partial \tilde{v}} & \frac{\partial v}{\partial \tilde{v}} \end{pmatrix} \begin{pmatrix} \partial_u \mathbf{r} \\ \partial_v \mathbf{r} \end{pmatrix}, \quad \begin{pmatrix} du & dv \end{pmatrix} = \begin{pmatrix} d\tilde{u} & d\tilde{v} \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial \tilde{u}} & \frac{\partial v}{\partial \tilde{u}} \\ \frac{\partial u}{\partial \tilde{v}} & \frac{\partial v}{\partial \tilde{v}} \end{pmatrix},$$

where we will denote the matrix

$$J := \begin{pmatrix} \frac{\partial u}{\partial \tilde{u}} & \frac{\partial v}{\partial \tilde{u}} \\ \frac{\partial u}{\partial \tilde{v}} & \frac{\partial v}{\partial \tilde{v}} \end{pmatrix}.$$

Hence we know that

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = \begin{pmatrix} \partial_{\tilde{u}} \mathbf{r} \\ \partial_{\tilde{v}} \mathbf{r} \end{pmatrix} \begin{pmatrix} \partial_{\tilde{u}} \mathbf{r} & \partial_{\tilde{v}} \mathbf{r} \end{pmatrix} = J \begin{pmatrix} \partial_u \mathbf{r} \\ \partial_v \mathbf{r} \end{pmatrix} \begin{pmatrix} \partial_u \mathbf{r} & \partial_v \mathbf{r} \end{pmatrix} J^T = J \begin{pmatrix} E & F \\ F & G \end{pmatrix} J^T,$$

therefore we have the computation

$$\begin{aligned}
\tilde{I} &= \tilde{E}(\mathrm{d}\tilde{u})^2 + 2\tilde{F}\mathrm{d}\tilde{u}\mathrm{d}\tilde{v} + \tilde{G}(\mathrm{d}\tilde{v})^2 \\
&= \begin{pmatrix} \mathrm{d}\tilde{u} & \mathrm{d}\tilde{v} \end{pmatrix} \begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} \begin{pmatrix} \mathrm{d}\tilde{u} \\ \mathrm{d}\tilde{v} \end{pmatrix} \\
&= \begin{pmatrix} \mathrm{d}\tilde{u} & \mathrm{d}\tilde{v} \end{pmatrix} J \begin{pmatrix} E & F \\ F & G \end{pmatrix} J^T \begin{pmatrix} \mathrm{d}\tilde{u} \\ \mathrm{d}\tilde{v} \end{pmatrix} \\
&= \begin{pmatrix} \mathrm{d}u & \mathrm{d}v \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \mathrm{d}u \\ \mathrm{d}v \end{pmatrix} \\
&= E(\mathrm{d}u)^2 + 2F\mathrm{d}u\mathrm{d}v + G(\mathrm{d}v)^2 \\
&= I,
\end{aligned}$$

which completes the proof. ■

• For a regular parametric surface  $S : \mathbf{r} = \mathbf{r}(u, v)$ , consider two tangent vectors at the same point

$$\mathrm{d}\mathbf{r}(u, v) = \partial_u \mathbf{r}(u, v)\mathrm{d}u + \partial_v \mathbf{r}(u, v)\mathrm{d}v, \quad \delta \mathbf{r}(u, v) = \partial_u \mathbf{r}(u, v)\delta u + \partial_v \mathbf{r}(u, v)\delta v,$$

the inner product of  $\mathrm{d}\mathbf{r}$  and  $\delta \mathbf{r}$  is

$$\begin{aligned}
\mathrm{d}\mathbf{r}(u, v) \cdot \delta \mathbf{r}(u, v) &= (\partial_u \mathbf{r}(u, v)\mathrm{d}u + \partial_v \mathbf{r}(u, v)\mathrm{d}v) \cdot (\partial_u \mathbf{r}(u, v)\delta u + \partial_v \mathbf{r}(u, v)\delta v) \\
&= (\partial_u \mathbf{r} \cdot \partial_u \mathbf{r})\mathrm{d}u\delta u + (\partial_u \mathbf{r} \cdot \partial_v \mathbf{r})(\mathrm{d}u\delta v + \delta u\mathrm{d}v) + (\partial_v \mathbf{r} \cdot \partial_v \mathbf{r})\mathrm{d}v\delta v \\
&= E\mathrm{d}u\delta u + F(\mathrm{d}u\delta v + \delta u\mathrm{d}v) + G\mathrm{d}v\delta v \\
&= \begin{pmatrix} \mathrm{d}u & \mathrm{d}v \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \delta u \\ \delta v \end{pmatrix}.
\end{aligned}$$

We adopt the notation  $I(\mathrm{d}\mathbf{r}, \delta \mathbf{r}) := \mathrm{d}\mathbf{r}(u, v) \cdot \delta \mathbf{r}(u, v) = \begin{pmatrix} \mathrm{d}u & \mathrm{d}v \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \delta u \\ \delta v \end{pmatrix}$ , then  $|\mathrm{d}\mathbf{r}|^2 =$

$$I(\mathrm{d}\mathbf{r}, \mathrm{d}\mathbf{r}) = \mathrm{d}\mathbf{r} \cdot \mathrm{d}\mathbf{r} = \begin{pmatrix} \mathrm{d}u & \mathrm{d}v \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \mathrm{d}u \\ \mathrm{d}v \end{pmatrix}.$$

• Let  $S : \mathbf{r} = \mathbf{r}(u, v)$  be a regular parametric surface, for a smooth curve  $C : \mathbf{r} = \mathbf{r}(t) \triangleq \mathbf{r}(u(t), v(t))$ ,  $t \in [a, b]$  on  $S$ , we have

$$\mathbf{r}'(t) = \frac{\mathrm{d}\mathbf{r}(u(t), v(t))}{\mathrm{d}t} = \partial_u \mathbf{r} \cdot \frac{\mathrm{d}u(t)}{\mathrm{d}t} + \partial_v \mathbf{r} \frac{\mathrm{d}v(t)}{\mathrm{d}t},$$

and we also get

$$\begin{aligned}
|\mathbf{r}'(t)| &= \sqrt{\left( \partial_u \mathbf{r} \cdot \frac{\mathrm{d}u(t)}{\mathrm{d}t} + \partial_v \mathbf{r} \frac{\mathrm{d}v(t)}{\mathrm{d}t} \right)^2} \\
&= \sqrt{E \left( \frac{\mathrm{d}u(t)}{\mathrm{d}t} \right)^2 + 2F \frac{\mathrm{d}u(t)}{\mathrm{d}t} \frac{\mathrm{d}v(t)}{\mathrm{d}t} + G \left( \frac{\mathrm{d}v(t)}{\mathrm{d}t} \right)^2},
\end{aligned}$$

which means that the arc-length of  $C$  is

$$\begin{aligned} L &= \int_a^b |\mathbf{r}'(t)| dt \\ &= \int_a^b \sqrt{E \left( \frac{du(t)}{dt} \right)^2 + 2F \frac{du(t)}{dt} \frac{dv(t)}{dt} + G \left( \frac{dv(t)}{dt} \right)^2} dt. \end{aligned}$$

Now suppose we are given a regular parametric surface  $S : \mathbf{r} = \mathbf{r}(u, v)$  defined on  $(u, v) \in D \subseteq \mathbb{R}^2$ .

Consider a small patch on the surface bounded by the parameter curves

$$u = u_0, \quad u = u_0 + \Delta u,$$

$$v = v_0, \quad v = v_0 + \Delta v,$$

where we may take  $\Delta u, \Delta v > 0$ . Its area is approximately equal to the area of the parallelogram on the tangent plane at  $(u_0, v_0)$  spanned by the vectors  $\{(\Delta u)\partial_u \mathbf{r}, (\Delta v)\partial_v \mathbf{r}\}$ , whose area is

$$\begin{aligned} |(\Delta u)\partial_u \mathbf{r} \times (\Delta v)\partial_v \mathbf{r}| &= |\partial_u \mathbf{r} \times \partial_v \mathbf{r}| \Delta u \Delta v \\ &= |\partial_u \mathbf{r}| \cdot |\partial_v \mathbf{r}| \cdot \sin \langle \partial_u \mathbf{r}, \partial_v \mathbf{r} \rangle \cdot \Delta u \Delta v \\ &= \sqrt{(|\partial_u \mathbf{r}|^2 \cdot |\partial_v \mathbf{r}|^2) (1 - \cos^2 \langle \partial_u \mathbf{r}, \partial_v \mathbf{r} \rangle)} \Delta u \Delta v \\ &= \sqrt{|\partial_u \mathbf{r}|^2 \cdot |\partial_v \mathbf{r}|^2 - (\partial_u \mathbf{r} \cdot \partial_v \mathbf{r})^2} \Delta u \Delta v \\ &= \sqrt{EG - F^2} \Delta u \Delta v. \end{aligned}$$

This leads to the following definition.

**Definition 2.** The *area element* of a regular parametric surface  $S : \mathbf{r} = \mathbf{r}(u, v)$ ,  $(u, v) \in D \subseteq \mathbb{R}^2$  is given by

$$dA := |\partial_u \mathbf{r} \times \partial_v \mathbf{r}| du dv = \sqrt{EG - F^2} du dv.$$

The area of  $S$  is given by

$$\mathcal{A}(S) = \iint_D dA = \iint_D \sqrt{EG - F^2} du dv.$$

**Remark:** Let  $u = u(\tilde{u}, \tilde{v})$ ,  $v = v(\tilde{u}, \tilde{v})$  be an admissible change of parameters, then from the theory of calculus we see that

$$\mathcal{A}(s) = \iint_D \sqrt{EG - F^2} du dv = \iint_{\tilde{D}} \sqrt{EG - F^2} \left| \frac{\partial(u, v)}{\partial(\tilde{u}, \tilde{v})} \right| d\tilde{u} d\tilde{v},$$

in the proof of Proposition 1, we have obtained

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = J \begin{pmatrix} E & F \\ F & G \end{pmatrix} J^T, \quad J := \begin{pmatrix} \frac{\partial u}{\partial \tilde{u}} & \frac{\partial v}{\partial \tilde{u}} \\ \frac{\partial u}{\partial \tilde{v}} & \frac{\partial v}{\partial \tilde{v}} \end{pmatrix}.$$

Thus we get

$$\begin{aligned} \tilde{E}\tilde{G} - \tilde{F}^2 &= \det \begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = \det \begin{pmatrix} E & F \\ F & G \end{pmatrix} (\det J)^2 \\ &= (EG - F^2) \left( \frac{\partial u}{\partial \tilde{u}} \frac{\partial v}{\partial \tilde{v}} - \frac{\partial u}{\partial \tilde{v}} \frac{\partial v}{\partial \tilde{u}} \right)^2 \\ &= \sqrt{EG - F^2} \left( \frac{\partial(u, v)}{\partial(\tilde{u}, \tilde{v})} \right)^2, \end{aligned}$$

i.e.,

$$\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} = \sqrt{EG - F^2} \left| \frac{\partial(u, v)}{\partial(\tilde{u}, \tilde{v})} \right|,$$

thus

$$\iint_D \sqrt{EG - F^2} du dv = \iint_{\tilde{D}} \sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} d\tilde{u} d\tilde{v},$$

which means  $\mathcal{A}(S)$  is independent of the choice of any admissible change of parameters and hence well-defined.